$$= -\sqrt{\frac{25-16}{25}}$$

$$= -\frac{3}{5}$$

$$\text{Sin B} = -\sqrt{1-\cos^2 B}$$

$$= -\sqrt{1-\frac{144}{169}}$$

$$= -\sqrt{\frac{169-144}{169}}$$

$$= -\frac{5}{13}$$

(i) $\cos(A+B) = \cos A\cos B - \sin A\sin B$

$$= \frac{4}{5}\times\frac{12}{13} - \left(\frac{-3}{5}\right)\times\left(\frac{-5}{13}\right)$$

$$= \frac{48}{65} - \frac{15}{65} = \frac{33}{65}$$

(ii) $\sin(A-B) = \sin A\cos B - \cos A\sin B$

$$= \left(\frac{-3}{5}\right)\left(\frac{12}{13}\right) - \left(\frac{4}{5}\right)\left(\frac{-5}{13}\right)$$

$$= \frac{-36}{65} + \frac{20}{65} = \frac{-16}{65}$$

Example 4.7

Find the values of each of the following trigonometric ratios.

(i) $\sin 15^\circ$ (ii) $\cos(-105^\circ)$
(iii) $\tan 75^\circ$ (iv) $-\sec 165^\circ$

Solution

(i) $\sin 15^\circ = \sin(45^\circ - 30^\circ)$

$$= \sin 45^\circ\cos 30^\circ - \cos 45^\circ\sin 30^\circ$$

$$= \frac{1}{\sqrt{2}}\times\frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}}\times\frac{1}{2} = \frac{\sqrt{3}-1}{2\sqrt{2}}$$

(ii) $\cos(-105^\circ)$

$= \cos 105^\circ$

$= \cos(60^\circ + 45^\circ)$

$= \cos 60^\circ\cos 45^\circ - \sin 60^\circ\sin 45^\circ$

$$= \frac{1}{2}\times\frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2}\times\frac{1}{\sqrt{2}} = \frac{1-\sqrt{3}}{2\sqrt{2}}$$

(iii) $\tan 75^\circ$

$= \tan(45^\circ + 30^\circ)$

$$= \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ\tan 30^\circ}$$

$$= \frac{1+\frac{1}{\sqrt{3}}}{1-\frac{1}{\sqrt{3}}} = \frac{\sqrt{3}+1}{\sqrt{3}-1} = 2+\sqrt{3}$$

(iv) $\cos 165^\circ = \cos(180^\circ - 15^\circ)$

$= -\cos 15^\circ$

$= -\cos(60^\circ - 45^\circ)$

$= -(\cos 60^\circ\cos 45^\circ + \sin 60^\circ\sin 45^\circ)$

$$= -\left(\frac{1}{2}\times\frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2}\times\frac{1}{\sqrt{2}}\right)$$

$$= -\left(\frac{1+\sqrt{3}}{2\sqrt{2}}\right)$$

$$\therefore -\sec 165^\circ = \frac{2\sqrt{2}}{1+\sqrt{3}}$$

$$= \frac{2\sqrt{2}}{1+\sqrt{3}}\times\frac{1-\sqrt{3}}{1-\sqrt{3}}$$

$$= \sqrt{2}(\sqrt{3}-1)$$

Example 4.8

If $\tan A = m\tan B$, prove that $\frac{\sin(A+B)}{\sin(A-B)} = \frac{m+1}{m-1}$



Componendo and dividendo rule:

If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

Solution

Given $\tan A = m\tan B$

$$\frac{\sin A}{\cos A} = m\frac{\sin B}{\cos B}$$

$$\frac{\sin A\cos B}{\cos A\sin B} = m$$

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